

Università degli Studi di Cagliari

Corso di Laurea Magistrale
Economia, Finanza e Analisi dei Dati

Problem Set 2

Quantitative Macroeconomics

Taxation, automation and calibration

Group 7

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A.A. 2026/2027

India: 1980–2019 Sri Lanka: 1983–2001

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Conventions

Exercise 1 compares steady states. We use $\beta = 0.96$, $\delta = 0.08$, $\psi = 1.5$, $\varphi = 0.8$ and $\theta \in \{0.33, 0.40\}$; the code calls φ `phi`. There is no consumption tax. Welfare is utility per period in the stationary allocation; consumption-equivalent changes use the formula specified in the assignment. N is a fraction of available time, $k = K/N$, and r in Exercise 1 is the marginal product of capital before depreciation and taxes. Negative log-utility levels do not mean that consumption or output is negative.

For Exercise 2, s is a nominal share, p a relative price and w thousand constant 2021 PPP dollars per person per year. The calibration uses only the first and last years of the assigned windows. Prices are normalized separately to one in each window's first year.

1 Exercise 1: Tax labor, or tax the robots?

We compare steady states of the model in Lectures 5b–5d. Throughout this exercise,

$$\beta = 0.96, \qquad \delta = 0.08, \qquad \psi = 1.5, \qquad \varphi = 0.8,$$

and the initial capital share is $\theta = 0.33$. We use *period* utility

$$U = \varphi[\ln C + \psi \ln(1 - N)] + (1 - \varphi) \ln G.$$

Multiplying this expression by $1/(1 - \beta)$ would give stationary lifetime utility, but the welfare comparison specified in the assignment uses the period expression above. Tax rates in the text and tables are percentages unless written as decimals.

1.1 Part A - One household owns everything

(a) The household's problem

The representative household owns capital and supplies labor. Taking prices, tax rates and the public good as given, its Lagrangian is

The representative household owns the aggregate capital stock and supplies labor elastically. Preferences over infinite sequences of consumption $\{C_t\}_{t=0}^{\infty}$, labor hours $\{N_t\}_{t=0}^{\infty}$, and the public good $\{G_t\}_{t=0}^{\infty}$ are represented by the intertemporal utility function:

$$\mathcal{U}_0 = \sum_{t=0}^{\infty} \beta^t \{\phi [\ln C_t + \psi \ln(1 - N_t)] + (1 - \phi) \ln G_t\} \tag{1}$$

where β is the subjective discount factor, ψ governs the relative weight of leisure, and the parameter ϕ dictates the weight assigned to private utility versus the public good G_t.

Taking market prices $\{w_t, r_t\}$, proportional tax rates on labor τ_n and capital τ_k, and the sequence of public goods $\{G_t\}$ as given, the household maximizes lifetime utility subject to the sequence of period budget constraints:

$$C_t + K_{t+1} = (1 - \tau_n) w_t N_t + [1 - \delta + (1 - \tau_k) r_t] K_t \quad \forall t \geq 0 \tag{2}$$

where δ is the physical capital depreciation rate and K_0 is historically given.

The Lagrangian and the three first-order conditions are derived in Appendix A.1.

Equilibrium Equations Substituting condition (28) into conditions (29) and (30) eliminates the multiplier μ_t and gives the two key equations:

- **Euler Equation:**

$$\frac{C_{t+1}}{C_t} = \beta\left[1 - \delta + (1 - \tau_k)r_{t+1}\right] \tag{3}$$

- **Intratemporal Consumption–Leisure Choice:**

$$\frac{\psi C_t}{1 - N_t} = (1 - \tau_n)w_t \tag{4}$$

Equation (3) is the standard Euler equation: it simply balances the trade-off between consuming today versus saving for tomorrow. If the household gives up one unit of consumption today, tomorrow it receives that unit back plus the net return on capital, after paying depreciation δ and capital tax τ_k.

Equation (4) determines the daily labor supply: it equates the value of an hour of free time (the marginal rate of substitution between leisure and consumption) to what the household actually earns by working, which is the after-tax wage $(1 - \tau_n)w_t$.

Two details are worth noting here:

- First, the weight ϕ cancels out of both optimality conditions because it multiplies private consumption and leisure in exactly the same way.
- Second, government spending G_t does not show up in either equation. Since utility is additively separable, the household takes G_t as given from the government: an individual worker decides their hours and savings without taking into account that their own taxes help fund public services.

(b) The steady state

We start from the Euler equation derived in (a), $\frac{1}{C_t} = \beta\frac{1-\delta+(1-\tau_k)r_{t+1}}{C_{t+1}}$. For compactness, define

$$q \equiv \frac{1}{\beta} - 1 = \frac{1}{24}, \qquad a \equiv q + \delta = 0.1216667 \tag{5}$$

In a steady state, (3) implies:

$$(1 - \tau_k)r = a \tag{6}$$

Using $r = \theta k^{\theta-1}$, we obtain:

$$k = \left[\frac{\theta(1 - \tau_k)}{a}\right]^{1/(1-\theta)} \tag{7}$$

This equation gives us the steady-state capital–labor ratio. The main takeaway is that k depends directly on the capital tax τ_k, but the labor tax τ_n does not appear here at all. Taxing capital makes investment less rewarding, which lowers the amount of machines per worker. However, even though τ_n does not change k, it will still affect total capital $K = k \cdot N$ because it changes how many hours people choose to work (N).

The steady-state household budget is:

$$C = \left[(1 - \tau_n)w + ((1 - \tau_k)r - \delta)k\right] N \tag{8}$$

Since $(1 - \tau_k)r - \delta = q$, define $x_k \equiv qk$ and write:

$$C = \left[(1 - \tau_n)w + x_k\right] N \tag{9}$$

Here x_k is simply the net capital income earned per hour worked, after paying for depreciation. This equation shows that the household finances its consumption through two sources of income per hour: after-tax labor earnings $(1-\tau_n)w$, and net capital returns x_k.

Since $w = (1-\theta)k^\theta$, the ratio between capital income and labor income is:

$$\frac{x_k}{w} = \frac{qk^{1-\theta}}{1-\theta} = \frac{q\theta}{(1-\theta)a}(1-\tau_k) \tag{10}$$

Let $z \equiv x_k/w$ and substitute (9) into (4):

$$\psi\left[(1-\tau_n)+z\right]N = (1-\tau_n)(1-N) \tag{11}$$

Solving for hours gives:

$$N = \frac{1-\tau_n}{(1-\tau_n)(1+\psi)+\psi z} \tag{12}$$

With both taxes equal to zero ($\tau_k = 0, \tau_n = 0$), this becomes the formula from Lecture 5d:

$$N_0 = \frac{1}{1+\psi+\psi\frac{q\theta}{(1-\theta)a}} = 0.3632380 \tag{13}$$

The intuition behind this result comes down to wealth:

- If the household owned zero capital ($z = 0$), it would rely solely on its salary, and hours would be fixed at $N = \frac{1}{1+\psi} = \frac{1}{1+1.5} = 0.40$.
- However, because the representative household also owns the capital stock, it receives extra non-labor income ($x_k > 0$). Being richer through capital generates an income effect: the household buys more leisure time and ends up working fewer hours ($0.3632 < 0.40$). In short, a household that collects returns on capital can afford to work less than a worker who depends entirely on labor income.

(c) Taxes and hours

Before differentiating the steady-state allocation, we can anticipate how each tax instrument affects labor supply. A labor tax directly reduces the take-home pay associated with an additional hour of work, making leisure relatively cheaper and reducing the incentive to supply effort. In contrast, a capital tax reduces the accumulation of assets and depresses non-labor income, which usually forces households to work longer hours to compensate for the loss in wealth.

To evaluate these mechanisms rigorously and account for general-equilibrium wage feedback, we differentiate the steady-state equation for hours worked. Defining the auxiliary parameters $A \equiv \frac{q\theta}{(1-\theta)a}$, $s \equiv 1-\tau_n$, and $D \equiv (1+\psi)s + \psi A(1-\tau_k)$, we can write hours compactly as $N = s/D$. Direct differentiation with respect to both tax rates yields:

$$\frac{\partial N}{\partial \tau_n} = -\frac{\psi A(1-\tau_k)}{D^2} < 0 \tag{14}$$

$$\frac{\partial N}{\partial \tau_k} = \frac{s\psi A}{D^2} > 0 \tag{15}$$

Both derivatives preserve strict signs for any tax rates below one hundred percent, confirming our initial intuition.

The negative response of hours to the labor tax reflects the standard intratemporal substitution effect: when work is taxed, the opportunity cost of leisure falls and households shift away from

market production. The positive response to the capital tax, however, is guided by a wealth channel. By discouraging investment, the capital tax lowers the capital–labor ratio k as well as the marginal product of labor, which would typically encourage households to take more leisure. Nevertheless, the tax hits net capital earnings per hour worked, x_k, proportionally harder than the wage itself, causing the ratio x_k/w to contract. Because the household experiences a sharp drop in passive non-labor income, it feels considerably poorer, and this negative income effect completely dominates the wage reduction, compelling the representative household to work more.

In the extreme limit where the capital tax approaches one hundred percent ($\tau_k \to 1$), capital income per hour completely dries up and the ratio x_k/w tends to zero. As a result, steady-state hours converge to the baseline fraction $N \to \frac{1}{1+\psi} = 0.40$, which corresponds to the labor supply of an agent with no non-labor wealth. This boundary case represents a purely mathematical limit rather than a desirable economic state: pushing τ_k to unity completely collapses the capital stock ($k \to 0$), destroying production, wages, consumption, and public revenues, which drives household utility down to negative infinity.

(d) Numerical tax experiments

For each tax rate in $\{0.001, 0.002, \ldots, 0.900\}$, we compute k, w, x_k, N, followed by

$$C = [(1-\tau_n)w + qk]N, \qquad G = [\tau_k\theta + \tau_n(1-\theta)]k^\theta N,$$

and period utility U. We first set $\tau_n = 0$ and then set $\tau_k = 0$. The resulting curves appear in Figure 1, and the selected rates are reported in Table 1. Rates are selected on the requested grid.

At $\theta = 0.33$, the optimal capital tax is 26.3% and hours are 0.372235. The optimal labor tax is 22.3% and hours are 0.353903. Revenue is maximized at much higher rates: 68.4% for the capital tax and 76.7% for the labor tax. Maximizing revenue therefore does not maximize welfare: high taxes reduce private consumption and, eventually, the tax base.

Table 1: Single-tax optima and revenue-maximizing rates.

θ	Tax	Welfare rate (%)	N	C	G	U	Revenue rate (%)
0.33	Capital only	26.3	0.37224	0.39440	0.04544	-1.92128	68.4
0.33	Labor only	22.3	0.35390	0.36656	0.08644	-1.81672	76.7
0.40	Capital only	20.9	0.36089	0.48344	0.05706	-1.69143	61.8
0.40	Labor only	22.9	0.33965	0.45028	0.10318	-1.59055	74.2

All optima use the assigned 0.001 grid. Utility is per period. Hours are a fraction of the time endowment, not weekly hours.

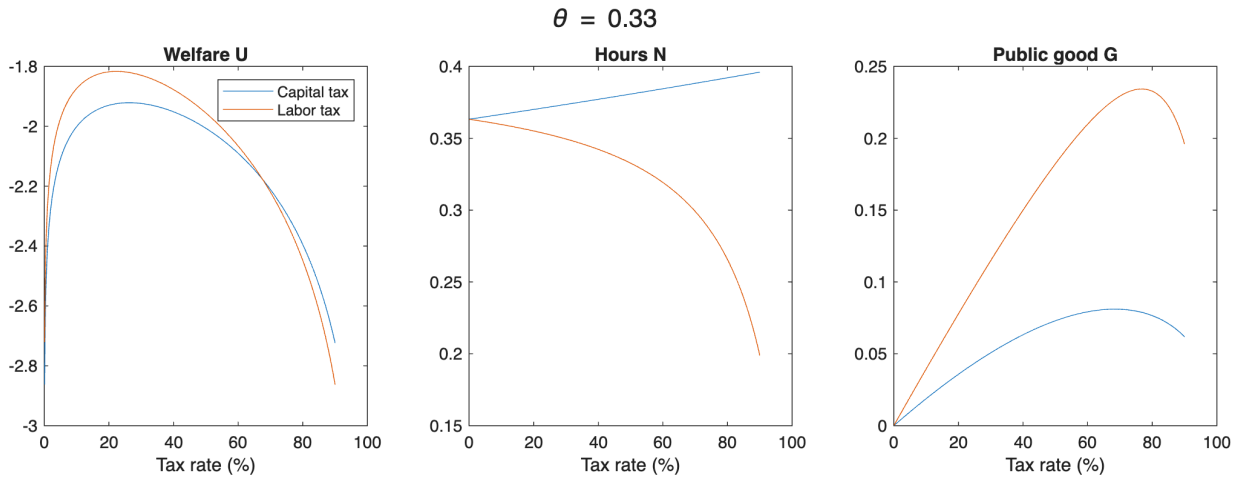


Figure 1: One tax at a time, $\theta = 0.33$. Panels (a), (b) and (c) show period welfare, hours and the public good over the requested tax grid. Each curve sets the other tax to zero.

As a self-check, we repeat the capital-tax calculation fixing hours at $N_0 = 0.363238$. The optimum is 25.6%, reproducing the representative household's fixed-hours experiment in Lecture 5c on the PS2 grid. Using Lecture 5c's original 200-point grid gives 25.7286% instead; rerunning both calculations on that same grid gives exactly the same maximizing rate. This small difference is only grid resolution, not a different model.

Appendix A.2 shows that, for a given capital tax, utility rises with hours as long as $N < 0.4545$.

Because hours stay below 0.40 and rise whenever we increase τ_k, letting hours respond gives an extra boost to the economy. Working more hours expands total production, consumption, and the tax base for the public good. The government realizes that taxing capital gives households an incentive to work longer, which partly offsets the loss of capital. This extra benefit nudges the optimal tax rate slightly higher, moving it from 25.6% up to 26.3%.

(e) Which tax, and what if both are available?

To compare the two optima, we use the assignment's consumption-equivalent measure

$$g = \exp\left(\frac{U_n^* - U_k^*}{\varphi}\right) - 1. \tag{16}$$

Increasing consumption by a factor $1+g$ raises period utility by $\varphi \ln(1+g)$, holding leisure and the public good fixed. Thus g is compensation needed under the capital-tax optimum to reach welfare under the labor-tax optimum.

At the initial capital share,

$$U_n^* = -1.8167190, \qquad U_k^* = -1.9212814, \qquad g = 0.1396292.$$

The labor tax is preferred by 13.9629% in consumption-equivalent terms. This does not mean that actual consumption is 13.9629% higher: private consumption is lower under the labor-tax optimum, but the household enjoys more leisure and a larger public good. Table 2 reports the welfare comparisons.

Table 2: Consumption-equivalent welfare comparisons (percent).

θ	HH: labor/capital	HH: cost of 25%	Worker: labor/capital	Capitalist: same switch
0.33	13.9629	8.0246	7.4516	88.8543
0.40	13.4400	11.3273	6.2989	72.7357

The first, third and fourth columns show the consumption increase required under the worse policy to match the better policy. The 25% column compares the overall optimum with the best labor tax conditional on $\tau_k = 0.25$.

Next, for each $\tau_k \in \{0, 0.05, \ldots, 0.40\}$, we choose the best labor tax on the same 0.001 grid. Table 3 shows the initial economy and Table 4 reports the automation comparison used below. Figure 2 plots the highest attainable utility at each background capital tax.

Table 3: Best labor tax for each capital tax, $\theta = 0.33$.

τ_k (%)	τ_n^* (%)	U^*	N	C	G
0.0	22.3	-1.816719	0.35390	0.36656	0.08644
5.0	20.3	-1.825259	0.35694	0.36489	0.08677
10.0	18.2	-1.835553	0.35992	0.36297	0.08655
15.0	16.1	-1.847736	0.36280	0.36032	0.08615
20.0	14.1	-1.861974	0.36555	0.35652	0.08591
25.0	12.0	-1.878470	0.36824	0.35231	0.08510
30.0	9.9	-1.897474	0.37084	0.34722	0.08408
35.0	7.9	-1.919296	0.37333	0.34085	0.08314
40.0	5.8	-1.944325	0.37578	0.33384	0.08161

The labor-tax search is 0.001:0.001:0.900. Every row recalculates the capital stock per worker and the equilibrium wage.

Table 4: Best labor tax for each capital tax, $\theta = 0.40$.

τ_k (%)	τ_n^* (%)	U^*	N	C	G
0.0	22.9	-1.590550	0.33965	0.45028	0.10318
5.0	20.1	-1.602208	0.34398	0.44799	0.10334
10.0	17.4	-1.616440	0.34805	0.44396	0.10358
15.0	14.6	-1.633403	0.35201	0.43916	0.10308
20.0	11.9	-1.653303	0.35575	0.43259	0.10263
25.0	9.1	-1.676394	0.35938	0.42513	0.10141
30.0	6.3	-1.702999	0.36287	0.41625	0.09981
35.0	3.5	-1.733524	0.36621	0.40588	0.09782
40.0	0.7	-1.768485	0.36942	0.39395	0.09541

The labor-tax search is 0.001:0.001:0.900. Every row recalculates the capital stock per worker and the equilibrium wage.

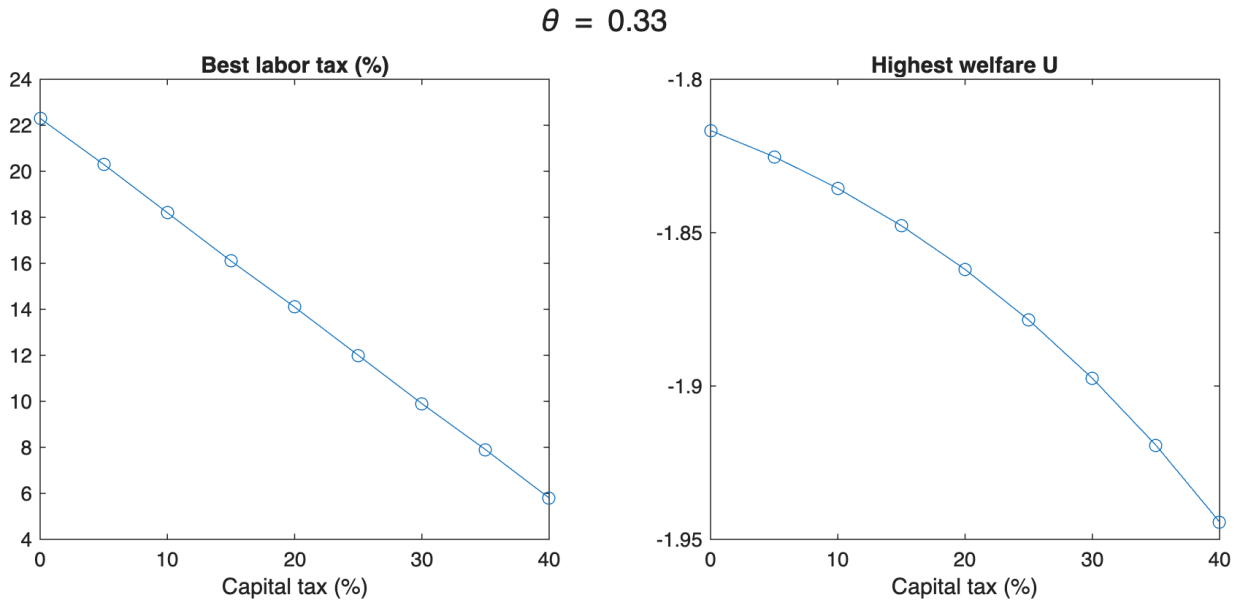


Figure 2: Joint taxation. At each background capital tax, the labor tax is chosen to maximize period welfare on the requested grid.

The best initial pair is

$$(\tau_k, \tau_n) = (0, 0.223).$$

If the capital tax must instead remain at 25%, the best labor tax is 12.0% and the highest period utility falls to -1.8784703. The assignment's compensation measure is

$$\exp\left(\frac{-1.8167190 - (-1.8784703)}{0.8}\right) - 1 = 0.0802463.$$

Therefore a household in the best equilibrium with a 25% capital tax would need 8.0246% more consumption to match the best joint-tax outcome.

Lecture 5c required a positive capital tax because that was the available way to finance the valued public good. Here the government can also tax labor. A capital tax permanently lowers the capital–labor ratio, productivity and wages, whereas a labor tax leaves k unchanged. In the joint grid the government prefers to finance G entirely through labor taxation, despite its effect on hours.

This is a comparison of steady states. It omits the transition: existing capital can yield revenue before investment and the capital stock fully respond, and moving to a higher-capital allocation takes time. Ignoring those dynamics can overstate the case for a zero capital tax. The numerical conclusion concerns the instruments and ranges specified in the assignment.

1.2 Part B - A worker and a capitalist

(f) Hours

We follow Lecture 5c and consider two households. The capitalist owns capital and saves; the worker supplies labor and does not save. She owns no capital. Both value the public good. The worker's budget is $c_w = (1 - \tau_n)wN$. Substituting it into her objective leaves the following terms that depend on hours:

$$\varphi[\ln N + \psi \ln(1 - N)].$$

The first-order condition is

$$\frac{1}{N} - \frac{\psi}{1 - N} = 0,$$

so

$$N = \frac{1}{1+\psi} = 0.4. \tag{17}$$

This result holds when the net wage is positive. It depends on neither tax nor θ.

With logarithmic preferences, a proportional change in the net wage has equal and opposite income and substitution effects for a worker with no non-labor income. In Part A, capital income broke that cancellation. Here automation does not reduce hours because the worker does not receive the additional income from the robots. The contrast with Lecture 5d shows why ownership matters for labor supply.

(g) A capitalist who enjoys wealth

Only the capital tax is used in this part. Let

$$R_t \equiv 1 - \delta + (1 - \tau_k) r_t.$$

Appendix A.3 writes the capitalist's Lagrangian and derives the steady state. With $q_\gamma \equiv c^k/K = (1-\tau_k)r - \delta$, the steady-state return is

$$q_\gamma = \frac{q}{1+\gamma}, \qquad r = \frac{\delta + q/(1+\gamma)}{1-\tau_k}. \tag{18}$$

At $\gamma = 0$ this is Lecture 5c's return. When $\gamma > 0$, wealth supplies a direct utility return, so a lower financial return is sufficient and the steady-state capital stock is larger.

The worker receives no direct utility from wealth. The financial return available to her satisfies

$$\beta R = \beta \left(1 + \frac{q}{1+\gamma}\right) < 1 \qquad (\gamma > 0).$$

Starting from no capital, she therefore has no incentive to accumulate positive assets. At $\gamma = 0$, $\beta R = 1$ and constant labor income can finance constant consumption with zero assets.

We now derive preferred capital taxes. The steps are in Appendix A.3. The capitalist prefers

$$\tau_k^k = \frac{(1-\varphi)(1-\theta)}{1+\varphi\gamma}. \tag{19}$$

The worker prefers

$$\tau_k^w = \frac{(1-\varphi)(1-\theta)}{1-\varphi+\varphi\theta}. \tag{20}$$

At $\theta = 0.33$, the capitalist prefers 13.4% when $\gamma = 0$ and 9.5714% when $\gamma = 0.5$. The worker prefers 28.8793% in both cases. Table 5 also reports the automation values.

Table 5: Preferred capital taxes in the split-ownership economy.

θ	γ	Worker (%)	Capitalist (%)	N	βR
0.33	0.0	28.8793	13.4000	0.400	1.000000
0.33	0.5	28.8793	9.5714	0.400	0.986667
0.40	0.0	23.0769	12.0000	0.400	1.000000
0.40	0.5	23.0769	8.5714	0.400	0.986667

$R = 1 - \delta + (1-\tau_k)r$ is the worker's gross financial return. The additional wealth motive belongs only to the capitalist.

A larger γ makes the capitalist more concerned about the reduction in K, so he prefers a lower capital tax. It changes the level of k enjoyed by the worker but not the elasticity of k with respect to $1-\tau_k$; her preferred rate is unchanged. Endogenous labor changes none of these preferred rates because hours are constant in the two-person economy.

(h) Preferred labor taxes

For the rest of the exercise, set $\gamma = 0$. With $\tau_k = 0$, the labor tax changes neither k nor N, and thus neither w nor capitalist consumption. For the worker,

$$c_w = (1 - \tau_n)wN, \qquad G = \tau_n wN,$$

so

$$U_w(\tau_n) = \text{constant} + \varphi \ln(1 - \tau_n) + (1 - \varphi) \ln \tau_n.$$

The preferred labor tax is

$$\tau_n^w = 1 - \varphi = 0.20. \tag{21}$$

At this rate,

$$\frac{G}{c_w} = \frac{\tau_n^w}{1 - \tau_n^w} = \frac{1 - \varphi}{\varphi} = 0.25.$$

The worker wants one unit of the public good for every four units of private consumption.

The capitalist's private consumption is unaffected by τ_n, while G rises with it. He therefore prefers a labor tax approaching 100%. This extreme result follows from the worker's exactly wage-inelastic hours and the capitalist's absence of labor income: within this model, raising the labor tax has no cost for him. A literal 100% tax would make worker consumption zero, so the preferred endpoint should be understood as a limit.

(i) Which tax does the worker prefer?

We compare the worker at her preferred capital tax, $\tau_k = 28.8793\%$, with her at her preferred labor tax, $\tau_n = 20\%$. The other tax is zero in each case. Table 6 reports private consumption, capitalist consumption, public spending and worker welfare.

Table 6: Outcomes at the worker's preferred single-tax policies ($\gamma = 0$).

θ	Policy	c_w	c_k	G	U_w	U_k
0.33	Capital tax	0.37040	0.04444	0.05269	-1.99620	-3.07966
0.33	Labor tax	0.35048	0.07390	0.08762	-1.93870	-2.57101
0.40	Capital tax	0.44549	0.07824	0.06854	-1.79592	-2.57446
0.40	Labor tax	0.42452	0.12115	0.10613	-1.74706	-2.13719

At $\theta = 0.33$, the worker's period welfare rises from -1.9962003 under her preferred capital tax to -1.9387040 under her preferred labor tax. Therefore

$$g_w = \exp\left(\frac{-1.9387040 - (-1.9962003)}{0.8}\right) - 1 = 0.0745160.$$

She prefers taxing her own wage by 7.4516% in consumption-equivalent welfare.

Worker consumption falls from 0.370403 to 0.350479, a decline of 5.3790%, while G rises from 0.052687 to 0.087620, an increase of 66.3036%. Removing capital taxation preserves capital accumulation, productivity and the revenue base. The public-good improvement more than offsets the worker's private-consumption loss.

The capitalist's consumption rises from 0.044435 to 0.073897, also by 66.3036%, and he benefits from the same larger public good. Using his utility and the same consumption-equivalent formula, his gain is 88.8543%. The worker still prefers the labor tax, even though the capitalist gains much more.

1.3 Part C - Automation

(j) What moves?

We now change only the capital share and repeat the calculations. Table 7 collects the four rates requested in the assignment, while Figures 3 and 4 show the tax curves and the changes in preferred rates.

Table 7: Automation: the four requested optimal rates (percent).

Policy	$\theta = 0.33$	$\theta = 0.40$
Household: capital tax only	26.300	20.900
Household: labor tax only	22.300	22.900
Worker: capital tax only	28.879	23.077
Worker: labor tax only	20.000	20.000

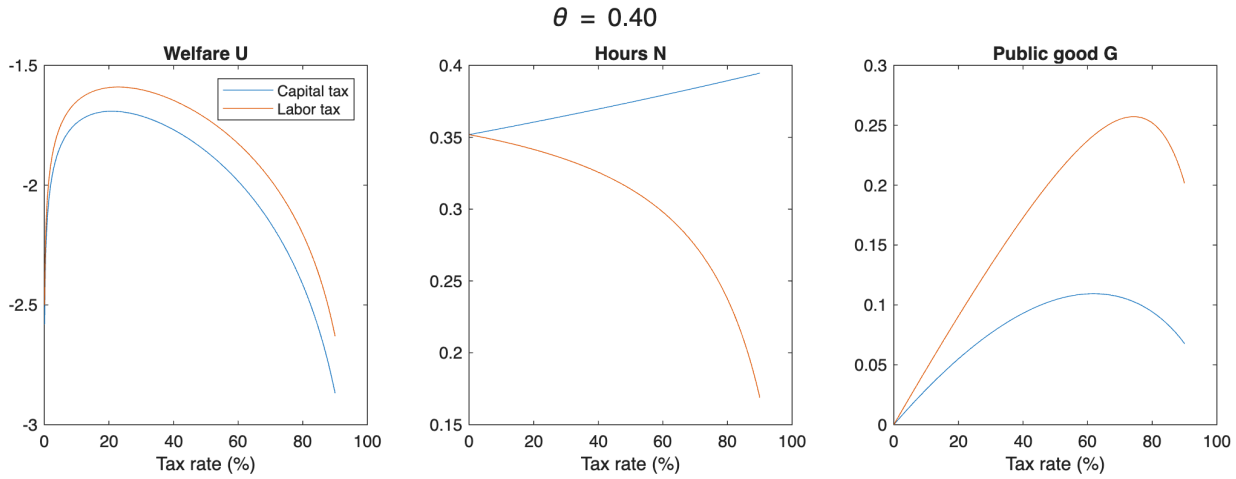


Figure 3: One tax at a time after automation, $\theta = 0.40$. Panels (a), (b) and (c) show period welfare, hours and the public good, setting the other tax to zero.

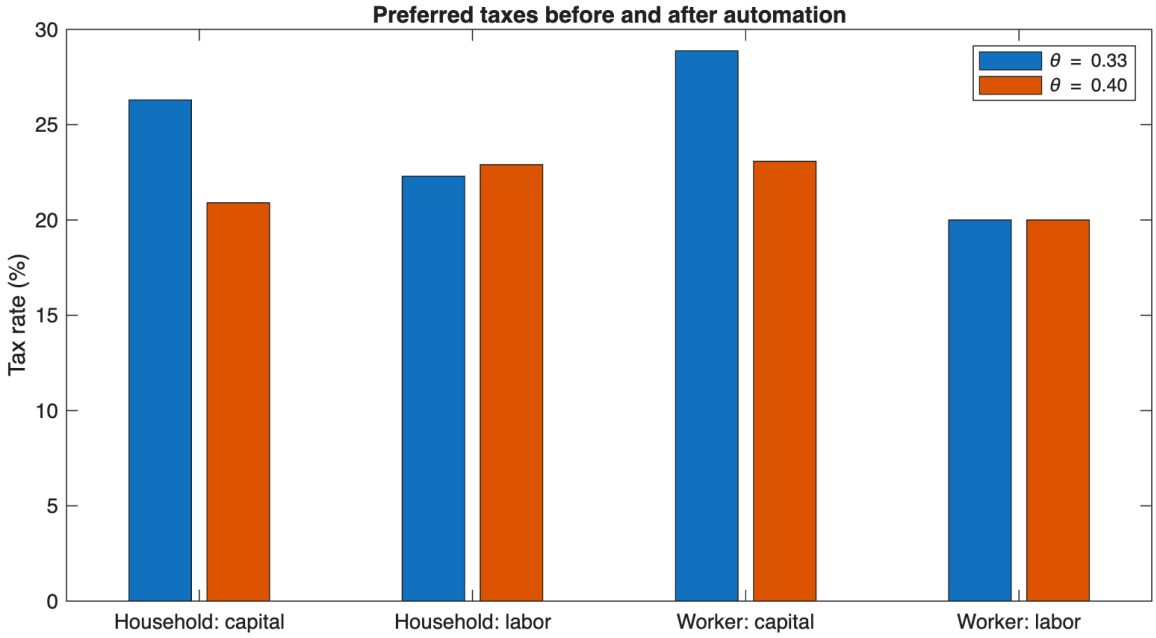


Figure 4: Preferred tax rates before and after automation. Household rates come from the 0.001 grid; worker rates are given by the analytical formulas.

The representative household's capital-only optimum falls from 26.3% to 20.9%, while its labor-only optimum rises slightly from 22.3% to 22.9%. Its hours at these new optima are 0.360895 and 0.339653, respectively. The worker's preferred capital tax falls from 28.8793% to 23.0769%, whereas her labor-tax preference remains 20%.

The worker's labor-tax formula depends only on the utility weights. In contrast, both the capital-tax formulas and the representative household's hours response depend on θ. In this calibration, a larger capital share makes the capital-accumulation cost of capital taxation more important. The representative household's optimal labor tax changes because capital income relative to wages, and hence the labor-supply response, changes.

Automation does not reverse the conclusion of part (e). Labor taxation still gives higher representative-household welfare, now by 13.4400% in consumption-equivalent terms. The best joint pair is $(0, 0.229)$. Keeping a capital tax of 25% instead gives a best labor tax of 9.1% and requires 11.3273% consumption compensation. The worker also continues to prefer the labor-tax option, by 6.2989%; her hours remain 0.4.

(k) A capital tax in the background

Return to the worker and fix τ_k while choosing τ_n. From the worker's labor supply and the capitalist's Euler equation, N, k and Y do not depend on τ_n. Since $rK/wN = \theta/(1-\theta)$,

$$c_w = wN(1-\tau_n), \qquad G = wN\left[\tau_n + \tau_k \frac{\theta}{1-\theta}\right].$$

Maximizing her welfare (Appendix A.4) gives

$$\tau_n^w = (1-\varphi) - \varphi\tau_k \frac{\theta}{1-\theta}. \tag{22}$$

At the interior solution, $G/c_w = (1-\varphi)/\varphi$ still holds: existing capital-tax revenue replaces part of the labor revenue needed to reach the desired public/private consumption ratio.

With $\tau_k = 0.25$, the preferred labor tax is 10.1493% at $\theta = 0.33$ and 6.6667% at $\theta = 0.40$, as shown in Table 8. Automation raises capital-tax revenue relative to the wage bill, so less labor taxation is needed.

Table 8: Worker's labor-tax preference with a background capital tax of 25%.

θ	τ_n^w (%)	Zero-tax threshold θ
0.33	10.1493	0.500
0.40	6.6667	0.500

Setting (22) equal to zero yields

$$\bar{\theta} = \frac{1-\varphi}{(1-\varphi)+\varphi\tau_k} = \frac{0.2}{0.2+0.8(0.25)} = 0.5.$$

For $\theta > 0.5$, the unconstrained optimum is a labor subsidy financed from capital-tax revenue. If labor taxes are restricted to be nonnegative, the worker instead chooses the corner $\tau_n = 0$.

(l) Back to the debate

Within this model, automation does not justify shifting additional tax burden from labor to capital: the representative household's capital-only optimum falls from 26.3% to 20.9%, and the joint search chooses a zero capital tax. The worker's preferred capital tax also falls, from 28.88% to 23.08%, while her preferred labor tax remains 20%. She prefers labor taxation by 7.45% in consumption-equivalent welfare before automation and 6.30% after it. With an existing 25% capital tax, however, automation increases its revenue relative to wages and lowers her preferred labor tax from 10.15% to 6.67%. The decisive assumption is that capital fully adjusts between steady states, so capital taxation permanently lowers investment, productivity and wages.

2 Exercise 2: Richer, or pricier?

The question is whether services gain a larger nominal share because income rises or because their relative price changes. We use the two-sector model of Lectures 3 and 10 and the numerical tools of Lecture 6b. Group 7 works on India, 1980–2019, and Sri Lanka, 1983–2001. The price of goods is the numeraire; goods combine agriculture and industry.

2.1 (a) Data, endpoints and the Pakistan check

We calculate the services share and the price of services relative to goods as

$$s_t = \frac{\text{srv_cd}_t}{\text{agr_cd}_t + \text{ind_cd}_t + \text{srv_cd}_t}, \tag{23}$$

$$p_t = \frac{\text{srv_cd}_t/\text{srv_kd}_t}{(\text{agr_cd}_t + \text{ind_cd}_t)/(\text{agr_kd}_t + \text{ind_kd}_t)}. \tag{24}$$

Current-price value added belongs in the nominal share. Constant-price series are used to obtain the implicit sector deflators and the real share, not as a substitute for current-price spending. For 1990 onwards, income is $w_t = \text{gdppc_ppp}_t/1000$. Before 1990, we extend the PPP level backwards using real GDP-per-person growth:

$$w_t = w_{1990}\frac{\text{gdppc_kd}_t}{\text{gdppc_kd}_{1990}}.$$

Thus all incomes remain in thousands of constant 2021 PPP dollars per person per year. We plot the full 1980–2019 histories, but use only the assigned windows for calibration and fit.

Table 9: Calibration inputs at the assigned endpoints.

Country	Year	s (%)	Raw p	Normalized p	w
India	1980	36.1373	1.3221	1.0000	1.5682
India	2019	54.7784	1.0086	0.7629	7.7022
Sri Lanka	1983	44.9841	0.7099	1.0000	3.7097
Sri Lanka	2001	53.1128	0.9112	1.2836	6.2300

w is thousand constant 2021 PPP dollars per person per year. The raw relative-price series has base 2015; the calibration divides it by its value in each country's first assigned year.

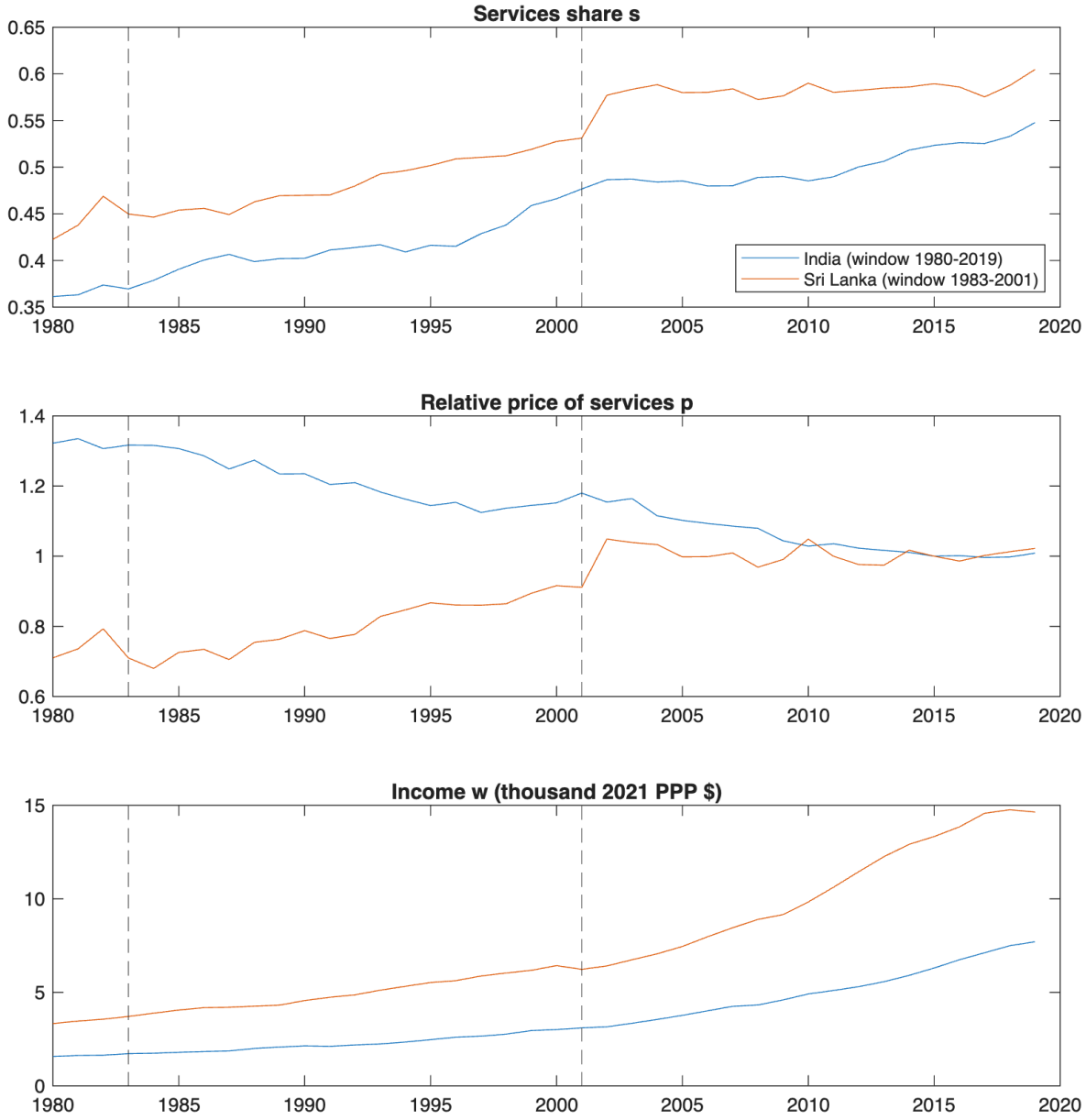


Figure 5: India and Sri Lanka, 1980–2019: (a) nominal services share, (b) raw relative price of services, and (c) PPP income per person. The dashed vertical lines mark the Sri Lanka window (1983–2001); the India window is the whole period.

India's services share rises by 18.64 percentage points while services become about 23.71% cheaper relative to goods. Sri Lanka's services share rises by 8.13 points while its relative price rises by 28.36%. Income rises in both countries.

The Pakistan exercise is a data-quality check required for every group, even though Pakistan is not one of our calibration countries.

Table 10: Pakistan: inspecting the 1999–2000 discontinuity.

Statistic	1999	2000	Change
Relative price p	0.9382	1.1847	23.3326 log points
Nominal services share (%)	49.2298	55.0159	5.7861 pp
Real services share (%)	50.8249	50.7948	-0.0301 pp

The price change is $100\Delta \ln p$, not a percentage-point change in a share. The corresponding ordinary price increase is 26.279%.

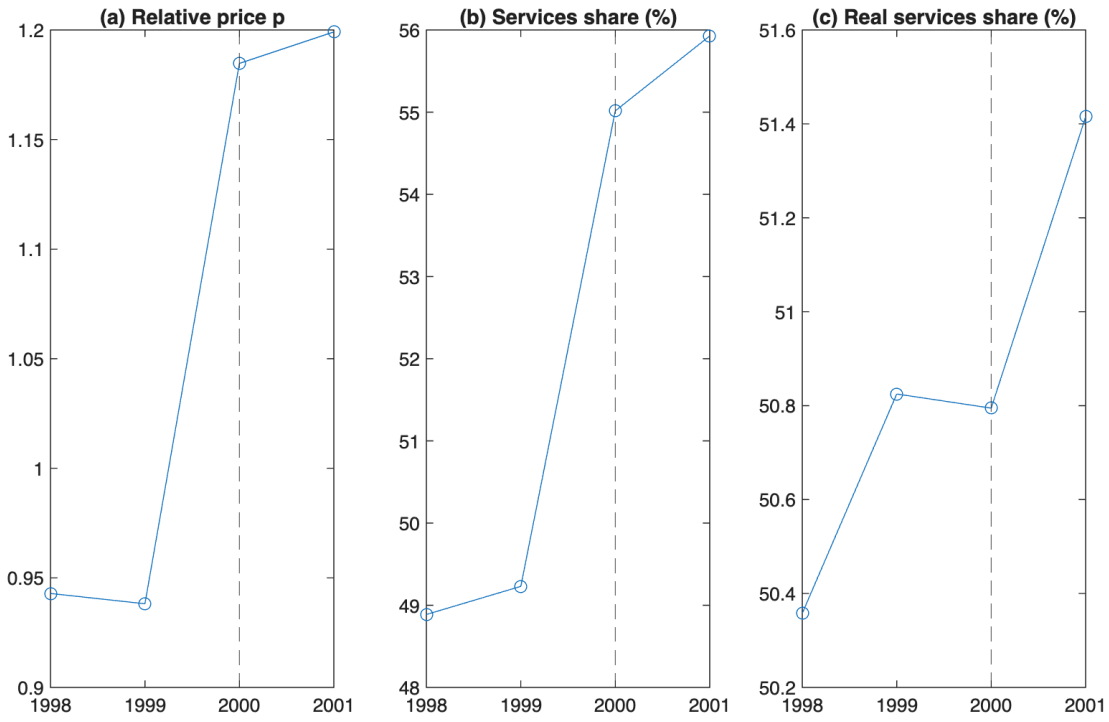


Figure 6: Pakistan around the 1999–2000 discontinuity: (a) relative services price, (b) nominal services share and (c) real services share. The vertical line marks 2000.

The nominal share and implied relative price jump, while the real services share barely changes. A change in the statistical basis, a deflator revision or a sectoral reclassification could produce this pattern; a large genuine relative-price change is another possibility. These aggregate series alone do not establish the cause. Calibrating mechanically to the year of a possible break could make the parameters explain a measurement discontinuity as if it were a change in household behaviour. This motivates respecting the windows supplied by the assignment.

2.2 (b) Deriving the services share

Let $x = c_m - \bar{m}$ and $I = w - \bar{m}$. The problem becomes a CES choice over x and c_s with budget $x + pc_s = I$. Appendix B.1 solves it step by step. The fraction of income spent on services is

$$\boxed{s = \frac{pc_s}{w} = H(p)\left(1 - \frac{\bar{m}}{w}\right), \qquad H(p) = \frac{1}{1 + [b/(1-b)]^{\varepsilon} p^{\varepsilon - 1}}.} \tag{25}$$

Here $\varepsilon = 1/(1-\rho)$. The corresponding goods demand is $c_m = \bar{m} + [1 - H(p)](w - \bar{m})$. A positive interior solution requires $p > 0$, $0 < b < 1$, $\rho < 1$ and $w > \bar{m} > 0$. At $\rho = 0$ the CES expression is interpreted by its Cobb–Douglas limit; $H(p) = 1 - b$, so the share formula remains well defined.

2.3 (c) The two forces and a prediction before calibration

Holding prices fixed, differentiate equation (25):

$$\frac{\partial s}{\partial w} = H(p)\frac{\bar{m}}{w^2} > 0.$$

The minimum goods requirement takes a smaller fraction of a richer household's income. More income is left for discretionary spending, and the services share rises towards $H(p)$. For the price effect,

$$H'(p) = (1 - \varepsilon)\frac{H(p)[1 - H(p)]}{p}, \qquad \frac{\partial s}{\partial p} = \left(1 - \frac{\bar{m}}{w}\right) H'(p).$$

Thus the nominal services share rises with its relative price if $\varepsilon < 1$, is unchanged if $\varepsilon = 1$, and falls if $\varepsilon > 1$. When substitution is difficult, the reduction in services quantities is too small to offset their higher unit price. When substitution is easy, households switch away sufficiently to reduce the spending share.

Before fitting parameters, India's large income increase and falling services price suggest an income-led rise: under complementarity the price force would work in the opposite direction. For Sri Lanka, rising income and rising prices would both raise the share under complementarity. A working prediction is that income contributes somewhat more, given its roughly 68% rise compared with a 28% price rise. This is a prediction to check, rather than a conclusion from comparing raw growth rates: the magnitudes also depend on $\bar{m}$ and ε.

2.4 (d) Four equations, four parameters

Each country has its own b_i, while ρ and $\bar{m}$ are common. With one country, the two endpoint targets would leave three unknowns: $b_i, \rho, \bar{m}$. Adding the second country supplies two equations and only one additional unknown, b_B. We therefore solve four endpoint equations for

$$\mathbf{x} = [b_A, b_B, \rho, \bar{m}].$$

Counting equations is necessary here, but does not by itself guarantee a unique economically valid solution. We normalize prices separately within each country: $\widetilde{p}_{i,t} = p_{i,t}/p_{i,1}$. Thus each country's first assigned price equals one.

The function `share_model` implements equation (25) with elementwise operations. The function `calib2` returns the four differences between model and data shares. The main script calls `fsolve` from the required starting point

$$[0.5, 0.5, 0, 0.25 \min(w_{A,1}, w_{B,1})].$$

Before using observed targets, we generate artificial shares at the specified parameter vector and recover it with the same solver.

Table 11: Synthetic-data recovery before the actual calibration.

Parameter	Artificial value	Recovered value
b_A	0.300000	0.3000
b_B	0.400000	0.4000
ρ	-1.500000	-1.5000
$\bar{m}$	0.800000	0.8000

The largest difference between the recovered and the artificial values is 7.9×10^{-7}.

The actual calibration also converges from the required first starting point: `exitflag=1`, with maximum endpoint residual below 10^{-9}. All parameter-domain checks pass, and subsistence expenditure is below income in every assigned year.

Table 12: Jointly calibrated parameters and implied subsistence expenditure.

Quantity	Value
b_A (India)	0.0958
b_B (Sri Lanka)	0.3129
ρ	-2.6739
$\varepsilon = 1/(1-\rho)$	0.2722
$\bar{m}$ (thousand PPP dollars/year)	0.6938
$1000\bar{m}/365$ (PPP dollars/day)	1.9009

Table 13: Subsistence-goods expenditure as a fraction of income.

Country	Year	$\bar{m}/w$	Percent of income
India	1980	0.4424	44.24
India	2019	0.0901	9.01
Sri Lanka	1983	0.1870	18.70
Sri Lanka	2001	0.1114	11.14

The low elasticity, about 0.272, implies that the price effect has the complementarity sign. The calibrated goods requirement is about \$1.90 per person per day in 2021 PPP units. This is below the \$3.00 benchmark specified by the assignment, at about 63.4% of that benchmark. The comparison is only a scale check: $\bar{m}$ is the model's goods requirement, not a full household poverty budget or an estimate of the proportion of people in poverty. Likewise, $\bar{m}/w$ is an expenditure fraction, not a poverty headcount.

2.5 (e) Testing the years in between

We keep the four fitted parameters fixed and insert each year's observed income and normalized price into equation (25). For country i, with n_i strictly interior years, we calculate

$$\mathrm{RMSE}_i = 100\sqrt{\frac{1}{n_i}\sum_{t\in\text{interior}}[s_{i,t}^{\text{model}} - s_{i,t}^{\text{data}}]^2}.$$

The factor 100 expresses the result in percentage points. The endpoints are excluded because the calibration already forces the model to match them.

Table 14: Model fit in years strictly between the calibrated endpoints.

Country	Window	Interior years	RMSE (pp)	Worst year	Largest error (pp)
India	1980–2019	38	2.1651	1996	4.3343
Sri Lanka	1983–2001	17	0.9787	1990	1.5115

A positive signed error means the model overpredicts the services share. The maximum-error search covers the assigned window; endpoints have essentially zero error.

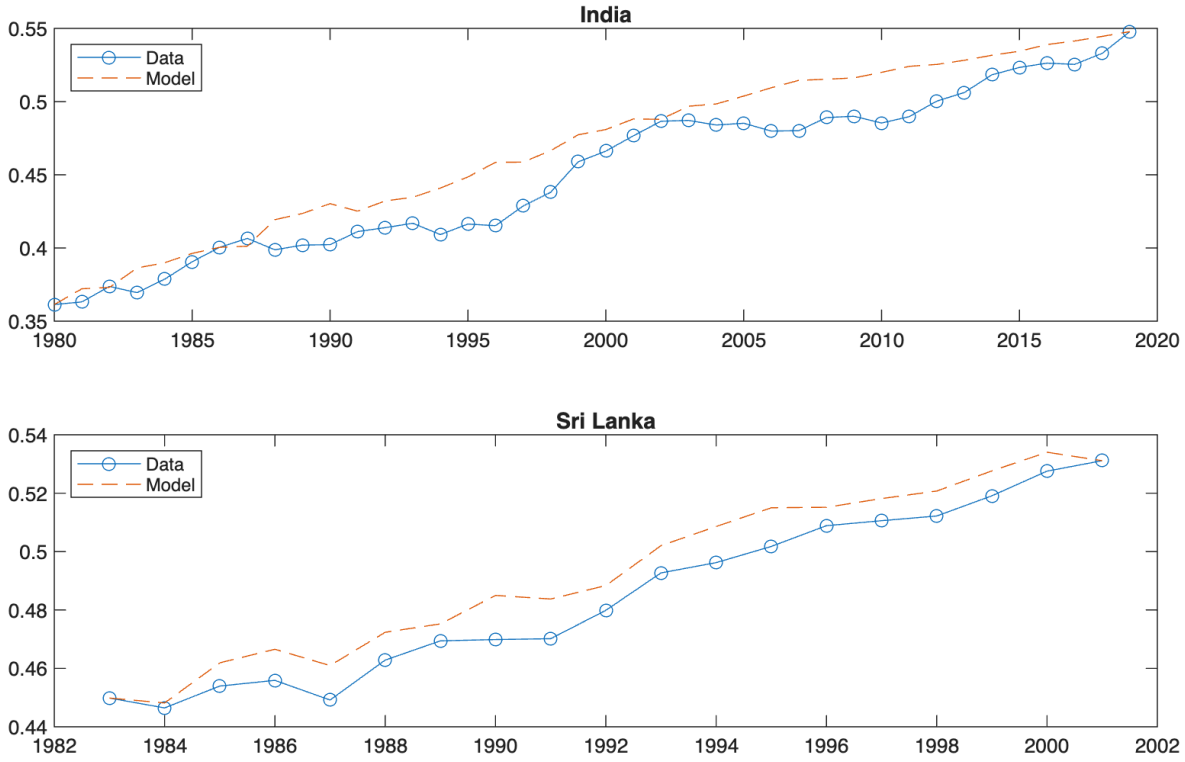


Figure 7: Observed and model services shares: (a) India, 1980–2019, and (b) Sri Lanka, 1983–2001. The first and the last year of each window are the calibration targets. The path between them is a prediction conditional on observed income and prices.

The model's largest overprediction is 4.33 percentage points in India in 1996, and 1.51 points in Sri Lanka in 1990. It captures the broad increase but does not explain every annual movement.

One reason is that household spending and domestic value added are different objects. Households can buy imported services or goods, while a country's production can be bought by foreign consumers, firms, investors or government. Trade and other final-demand components can therefore change the production share without the corresponding change in household preferences. GDP per person is also an approximation to household income. These measurement links help explain misses even with economically admissible parameters.

2.6 (f) Richer, or pricier?

Taking logs of equation (25) gives the exact model decomposition

$$\underbrace{\ln \frac{s_T}{s_1}}_{\text{total change}} = \underbrace{\ln \frac{H(p_T)}{H(p_1)}}_{\text{price force}} + \underbrace{\ln \frac{1-\bar{m}/w_T}{1-\bar{m}/w_1}}_{\text{income force}}. \tag{26}$$

Table 15: Exact decomposition of the rise of services: 100× log changes.

Country	Price force	Income force	Total	Income/total (%)
India	-7.3827	48.9798	41.5971	117.75
Sri Lanka	7.7117	8.8992	16.6108	53.57

These are log changes multiplied by 100, not percentage-point changes in the services share. A contribution may exceed 100% of the total when the other contribution is negative.

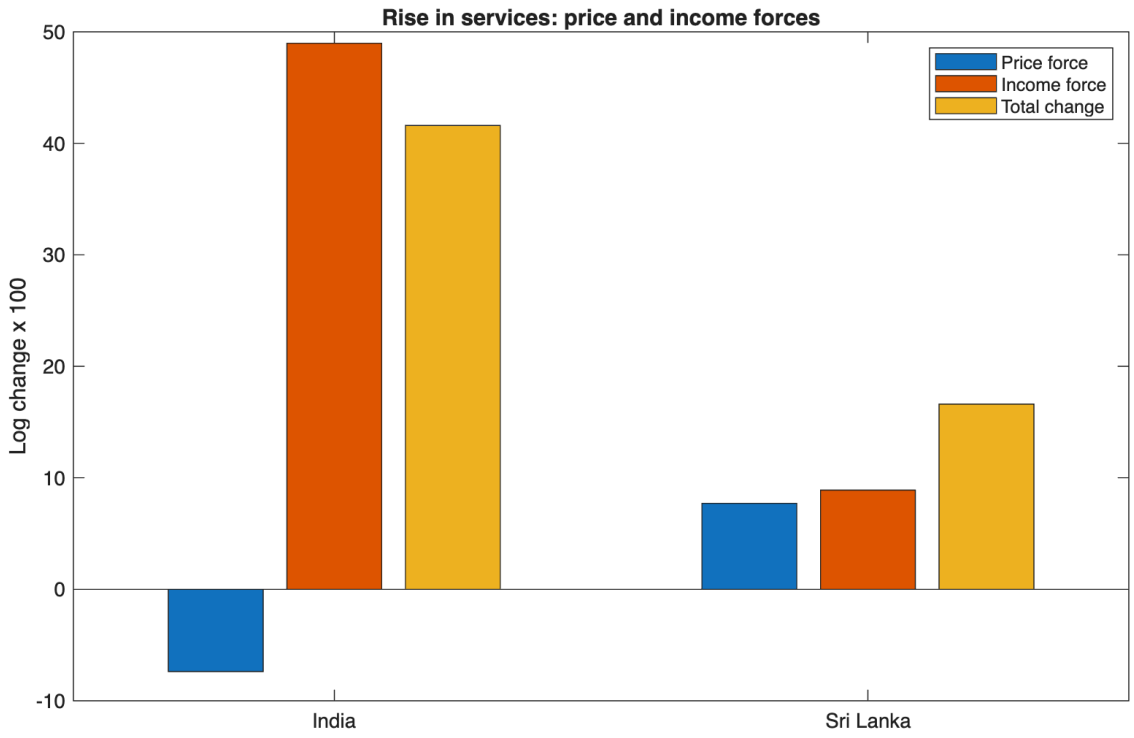


Figure 8: Price and income contributions to the rise of services, each expressed as a log change multiplied by 100. The two contributions sum to the total within each country.

India. The income contribution is 48.98 log points and the price contribution is −7.38, producing the observed total of 41.60. India becomes richer; cheaper services partly offset the rise in their nominal share because $\varepsilon < 1$. Income explains more than 100% of the net rise precisely because prices pull in the opposite direction. The income-led prediction in (c) is confirmed.

Sri Lanka. Both contributions are positive: income contributes 8.90 log points and prices 7.71, for a total of 16.61. The mechanisms are close in size, with income accounting for about 53.6% and prices 46.4%. Our working prediction that both matter and income contributes slightly more is confirmed.

If services were easy to substitute. With $\varepsilon > 1$, $H'(p) < 0$. Cheaper services would then generate a positive price force, because the quantity response would dominate the lower unit price. These calculations are log changes multiplied by 100; they are neither the change in the share measured in percentage points nor an empirical causal estimate of each force. The independent fit check is the comparison in (e).

A Derivations for Exercise 1

A.1 Part (a): Lagrangian and first-order conditions

The Lagrangian Function Following the convention of Lecture 5, the multiplier μ_t is defined outside the discount factor β^t:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \left\{\phi \left[\ln C_t + \psi \ln(1 - N_t)\right] + (1 - \phi) \ln G_t\right\} + \sum_{t=0}^{\infty} \mu_t \left[(1 - \tau_n) w_t N_t + (1 - \delta + (1 - \tau_k) r_t) K_t - C_t - K_{t+1}\right] \tag{27}$$

First-Order Conditions (FOCs) Differentiating with respect to the three choice variables $\{C_t, N_t, K_{t+1}\}$ yields:

$$\frac{\beta^t \phi}{C_t} = \mu_t \tag{28}$$

$$\frac{\beta^t \phi \psi}{1 - N_t} = \mu_t (1 - \tau_n) w_t \tag{29}$$

$$\mu_t = \mu_{t+1} \left[1 - \delta + (1 - \tau_k) r_{t+1}\right] \tag{30}$$

A.2 Part (d): why letting hours respond raises the optimal capital tax

The rise from 25.6% to 26.3% can also be seen directly. For a given capital tax, k is the same in both experiments, while C and G are proportional to N. Relative to fixed hours, the utility correction is:

$$\Delta U_N = \ln\left(\frac{N}{N_0}\right) + \phi\psi \ln\left(\frac{1 - N}{1 - N_0}\right) \tag{31}$$

Its derivative with respect to N is:

$$\frac{d\Delta U_N}{dN} = \frac{1}{N} - \frac{\phi\psi}{1 - N} > 0 \quad \text{when} \quad N < \frac{1}{1 + \phi\psi} = 0.454545 \tag{32}$$

A.3 Part (g): the capitalist's problem and the preferred capital taxes

The capitalist's Lagrangian is

$$\begin{aligned}\mathcal{L}_k = &\sum_{t=0}^{\infty} \beta^t \{\varphi[\ln c_t^k + \gamma \ln K_t] + (1 - \varphi) \ln G_t\} \\ &+ \sum_{t=0}^{\infty} \mu_t [K_t R_t - c_t^k - K_{t+1}].\end{aligned}$$

The relevant conditions are

$$\mu_t = \frac{\beta^t \varphi}{c_t^k}, \qquad \mu_t = \mu_{t+1} R_{t+1} + \frac{\beta^{t+1} \varphi \gamma}{K_{t+1}}.$$

The extra term is the direct utility benefit of owning tomorrow's capital. Combining the conditions,

$$\frac{1}{c_t^k} = \beta \left[\frac{R_{t+1}}{c_{t+1}^k} + \frac{\gamma}{K_{t+1}}\right]. \tag{33}$$

In a steady state, write

$$q_\gamma \equiv \frac{c^k}{K} = (1 - \tau_k) r - \delta.$$

Since $R = 1 + q_\gamma$, equation (33) becomes

$$\frac{1}{\beta} = 1 + (1+\gamma)q_\gamma.$$

Solving for q_γ gives (18).

Since $N = 0.4$,

$$k = \left[\frac{\theta(1-\tau_k)}{\delta + q/(1+\gamma)}\right]^{1/(1-\theta)}, \quad c^k = q_\gamma k N, \quad G = \tau_k \theta k^\theta N.$$

After collecting all terms that depend on the tax, the capitalist's utility is

$$U_k = \text{constant} + \frac{\varphi(1+\gamma) + (1-\varphi)\theta}{1-\theta}\ln(1-\tau_k) + (1-\varphi)\ln\tau_k.$$

For an expression $A\ln(1-\tau) + B\ln\tau$, the maximizing rate is $B/(A+B)$. Hence (19).

For the worker, $c_w = (1-\theta)k^\theta N$ and the leisure term is constant. Her tax-dependent utility is

$$U_w = \text{constant} + \theta\ln k + (1-\varphi)\ln\tau_k = \text{constant} + \frac{\theta}{1-\theta}\ln(1-\tau_k) + (1-\varphi)\ln\tau_k.$$

It follows (20).

A.4 Part (k): the worker's labor tax with a background capital tax

Ignoring constants, worker welfare is

$$U_w = \varphi\ln(1-\tau_n) + (1-\varphi)\ln\left[\tau_n + \tau_k\frac{\theta}{1-\theta}\right].$$

The first-order condition is

$$-\frac{\varphi}{1-\tau_n} + \frac{1-\varphi}{\tau_n + \tau_k\theta/(1-\theta)} = 0.$$

Solving gives (22).

B Derivations for Exercise 2

B.1 Part (b): the services share

Define $Z = bx^\rho + (1-b)c_s^\rho$. For $\rho \neq 0$, the Lagrangian is

$$\mathcal{L} = \frac{1}{\rho}\ln Z + \lambda[I - x - pc_s].$$

The two choice-variable conditions are

$$\frac{bx^{\rho-1}}{Z} = \lambda, \qquad \frac{(1-b)c_s^{\rho-1}}{Z} = \lambda p.$$

Dividing the first condition by the second eliminates both λ and Z:

$$\frac{b}{1-b}\left(\frac{x}{c_s}\right)^{\rho-1} = \frac{1}{p} \quad\Longrightarrow\quad \frac{x}{c_s} = \left(\frac{bp}{1-b}\right)^\varepsilon, \qquad \varepsilon = \frac{1}{1-\rho} > 0.$$

Substituting into $x + pc_s = I$ gives

$$c_s = \frac{w - \bar{m}}{p + [bp/(1-b)]^\varepsilon}.$$

Multiplying by p/w gives (25).